MARTIN DESSART

Postdoctoral Associate – Virginia Tech
1805 Shadow Lake Rd, Blacksburg, VA 24060, USA
Email: dessart@vt.edu
Phone: +1 (540) 824 9178
Website: martindessart.github.io



Research and Engineering Experience

10/2024 – Present
Department of Biochemistry
Virginia Tech - Blacksburg, VA

Postdoctoral Associate – Neuroethology
Investigating sensory integration and chronobiology in mosquitoes.
Mentor: Dr. Clément Vinauger

02/2021 – 02/2024
IRBI – UMR 7261, CNRS
University of Tours
Tours, France

Research in Neuroethology and Sensory Ecology
Examined the impact of pollutants on mosquito larvae cognition.
Assessed the role of larvae as bioindicators of freshwater ecosystems.
Supervisors: Prof. Claudio Lazzari & Dr. Fernando Guerrieri

07/2020 – 10/2020
CRCA, GEODE, CNRS
University Paul Sabatier
Toulouse, France

Research Internship
Collected flower reflectance in the field using spectrophotometry.
Modeled flower visual saliency using honeybee-vision models.
Supervisors: Dr. Sylvie Guillerme & Prof. Martin Giurfa

01/2020 – 07/2020
Animal Minds Lab
University of Auckland
Christchurch, New Zealand

Master's Final Project
Designed and conducted judgment bias tests on kea *Nestor notabilis*.
Analyzed keas vocalizations using Raven and Praat softwares.
Supervisors: Dr. Alexander Taylor & Dr. Amalia Bastos

04/2019 – 06/2019
UMR MARBEC, CNRS
University of Montpellier
Montpellier, France

Research Protocol Design / Assistant
Designed an experiment on fish gut methane emissions.
Performed field sampling and molecular analyses.
Supervisors: Dr. Jean-Christophe Auguet & Dr. M.C. Cheutin

09/2014 – 09/2017
Clemessy Aeronautics & Space
Toulouse, France

Apprentice Engineer
Designed and integrated automated cable-testing system.
Supervised electrical checks, managed client relationships.
Supervisor: Jean-Paul Herrgott

09/2016 – 02/2017
Auckland University
Auckland, New Zealand

Engineering Thesis
Analyzed wind-farm electricity generation.
Supervisors: Prof. Tek Tjing Lie & Dr. Gianni Goretti

05/2015 – 08/2015
TESEO – Clemessy Entity
Torino, Italy

Intern in Computing Design
Automated a maintenance procedure using VBA macros.
Conducted radar calibration in anechoic chamber.
Supervisor: Jean-Louis Menu

07/2013 – 08/2013
Tunisia Plastic Systems
Sousse, Tunisia

Intern in Production
Shorts assignments in production, logistics, and mechanical services.
Supervisor: Marcel Guillet

Education

02/2021 – 02/2024
University of Tours
Tours, France

PhD at Insect Biology Research Institute
Designed and conducted experiments in cognition, ecotoxicology.
Performed data analysis, programming, field work, teaching.

09/2018 – 09/2020
University of Strasbourg
Strasbourg, France

MSc (High Honours)
Studied neurobiology, ecophysiology, ethology, cognition.
Developed skills in statistical analyses and field work.

11/2017 – 08/2018
Udacity & Google School, online

Front-End Web Developer Nanodegree
Developed front-end designs and learned how to use frameworks.

09/2014 – 09/2017
Institut Catholique des Arts et Métiers
Toulouse, France

Engineer's Degree – Automation & Control Systems
Studied programming, project management, electronics engineering.

09/2012 – 09/2014
Institut Catholique des Arts et Métiers
Toulouse, France

Preparatory Class
Studied electrical engineering, automation, industrial organization.

Teaching and Mentoring

Mentoring

2024 – Present
Virginia Tech

Richard Rust (Graduate, **Neuroscience**)
Lily Smith (Senior, **Neuroscience**)
Jasmine Adams (Senior, **Neuroscience**)
Sydney Luff (Junior, **Biology & Entomology**)
Hayden Sunman (Freshman, **Biochemistry**)

03/2021 – 05/2021
University of Tours

Tristan Robineau (MSc Student, **Ecology & Evolution**)
Effect of light and turbidity on habituation of *Aedes aegypti* larvae.

06/2017 – 08/2027
Clemessy Aeronautics & Space

Eva Daoud (Engineering Student, **ESTACA Engineering School**)
Database design & conception

Teaching

2021 – 2023
University of Tours

University of Tours
Arthropod identification (32h, TP)
Neuroscience: conditioning and addiction (6h, TD)
Arthropod systematics (7.5h, TD)
Ethology (25h, TP)

Publications

Published

2025 **Dessart M.**, Lazzari C.R., Guerrieri F.J. (2025). *Acute and chronic sublethal chemical pollution affects locomotion, learning and memory in mosquito larvae.* Journal of Experimental Biology; jeb.250493. https://doi.org/10.1242/jeb.250493.

2024 **Dessart M.**, Lazzari C.R., Guerrieri F.J. (2024). *Habituation leads to short but not long-term memory formation in mosquito larvae.* Journal of Insect Physiology 155:104650. https://doi.org/10.1016/j.jinsphys.2024.104650

2024 **Dessart M.**, Aguiar J.M.R., Tabacchi E., Guillerme S., Giurfa M. (2024). *Color-advertising strategies of invasive plants through the bee eye.* Frontiers in Plant Science. 15, 1393204. https://doi.org/10.3389/fpls.2024.1393204

2023 **Dessart M.**, Piñeirúa M., Lazzari C.R., Guerrieri F.J. (2023). *Assessing learning in mosquito larvae using video-tracking.* Journal of Insect Physiology. 149, 104535. https://doi.org/10.1016/j.jinsphys.2023.104535

In preparation

a. **Dessart M.**, Luff S., Smith L., Sun H., Vinauger C. *Light-induced shift effect on mosquito circadian rhythms.* In preparation

b. **Dessart M.**, Rust R., Adams J., Smith L., Wynne N., Chandrasegaran K., Wolff G., Vinauger C. *Modulation of mosquitoes' responses to visual threats.* In preparation

c. Wynne N., Aronson H., **Dessart M.**, Luff S., Applebach E., Ajayi O., Chandrasegaran K., Benoit J., Lahondère C., Vinauger C. *Larval growing conditions affect the daily rhythms and thermal preferences of adult Aedes japonicus.* In preparation

d. **Dessart M.**, Lazzari C.R., Guerrieri F.J. *From field to lab, how environment shapes cognitive abilities in mosquito larvae?* In preparation

e. Cheutin M.C., Balland J., Roso E., **Dessart M.**, Rieuvilleneuve F., Villeger S., Bouvier T., Auguet J.C. *Sulfate-reduction prevents marine fishes from farting methane.* In preparation

Conference Presentations

2025 "Modulation of mosquitoes' responses to visual threats" – **Oral presentation** – 2025 Entomology, Portland, OR

2025 "The plasticity of mosquito circadian clock" – **Poster presentation** – Animal Behaviour Live 2025 Online Conference

2025 "How to jetlag a mosquito" – **Poster presentation** – 2025 CeZAP Infectious Disease Research Symposium - Virginia Tech, Blacksburg, VA

2024 "The cognition of mosquito larvae as a biological indicator of environmental quality" – **Oral presentation** – Club de Neuroéthologie – Société des Neurosciences – Albi, France

2024 "La cognition des larves de moustique comme indicateur biologique de la qualité des écosystèmes aquatique - **PhD Defense** - University of Tours, France

2023 "Cognitive abilities in mosquito larvae as indicators of environmental quality" – **Oral presentation** – Behaviour 2023 – Bielefeld, Germany

2023 "Chemical pollution affects the behaviour of mosquito larvae" – **Oral presentation** – DiPEE "Biodiversité et territoires: une recherche engagée" – CEBC Chizé, France

2023 "Chemical pollution affects the behaviour of mosquito larvae" – **Poster presentation** – 52ème Colloque of SFECA – Tours, France

2022 "Effect of two herbicides and one medical drug on locomotor activity and learning in *Aedes aegypti*" – **Oral presentation** – 34° Colloque Biotechnocentre – Centre Val de Loire, France

2022 "A tracking system to assess activity and learning in mosquito larvae" – **Poster presentation** – Ecology and Behaviour – 15th Edition – Strasbourg, France

2021 "Visual learning abilities in mosquito larvae are not impaired neither by light deprivation nor by turbidity" – **Oral presentation** – European PhD Network "Insect Science" – Florence, Italy

2021 "Invasive plant species through the bee eye: an analysis of flower coloration in the French Pyrenees" – **Oral presentation - Laboratory seminar** – IPHC UMR 7178 CNRS – Strasbourg, France

2021 "The influence of physico-chemical characteristics of aquatic ecosystems on prey–predator interactions. A behavioural-neuroanatomical approach" – **Oral presentation - Laboratory seminar** – IRBI UMR 7261 CNRS – University of Tours, France

2021 "Invasive plant species through the bee eye: an analysis of flower coloration in the French Pyrenees" – **Poster presentation** – 50ème Colloque of SFECA – Marseille, France

2020 "Effects of the warble call on optimism in Kea (*Nestor notabilis*)" – **Oral presentation - Laboratory seminar** – Research Center on Animal Cognition – University Paul Sabatier – Toulouse III

Academic and Professional Honors

2025 Travel grant to Entomology 2025 - Office of Research and Innovation, Virginia Tech

2024 Travel grant – French Society of Animal Behavior (SFECA), Albi, France

2024 Ph.D. obtained with High Honors, University of Tours, France

2023 Travel grant – Behaviour Conference, Bielefeld, Germany - University of Tours

2022 Regional finalist of My thesis in 180 seconds program, University of Tours, France

2021 3-years doctoral Fellowship, Region Centre Val de Loire, University of Tours, France

2019 Master's degree obtained with High Honors, University of Strasbourg, France

2019 Travel grant – Region Alsace GrandEst, Strasbourg, France

2017 Scholarship – 9-month training Udacity & Google online formation

2017	Engineering degree obtained with High Honors, Institut Catholique des Arts et Métiers, Toulouse, France

Service and Other Experience

2025 - Present	Board member of the Postdoc & *Research Faculty Committee* in the Department of Biochemistry, Virginia Tech
2025 - Present	Board member of the *Virginia Tech Postdoctoral Association*, Virginia Tech
2025 - Present	Review of manuscripts for *Scientific Reports* and *Ecology and Evolution*
2025 - Present	Member of the *Entomological Society of America*
2023 - Present	Member of the *French Society of Neuroethology*
2023 - Present	Member of the *International Society of Neuroethology*
2021 - Present	Member of the *French Society of Animal Behavior*
2023 - 2024	Member of the *German Society of Ethology (Ethologische Gesellschaft)*
2021 - 2024	Member of the *French Union for the Study of Social Insects (UIEIS)*
2020 - 2021	Contributed to the writing of a successfully funded 3-years PhD project (IPPY)
03/18 - 04/18	Volunteering: Assisted a vet and conducted interviews with farming communities. *Melindika Solidarity Association* - Kafue National Park, Zambia
01/18 - 02/18	Fieldwork assisting a PhD student with a protocol on cats to study their biological rhythms. *AVA Sanctuary* - Normandie, France
10/16 - 12/16	Volunteering: Assisted in the realisation of cognitive enrichment items for primates. *Auckland Zoo* - Auckland, New Zealand
2014 – 2017	Volunteering: Activity leader in France, Morocco, and Madagascar. *Terres en Mêlées Solidarity Association* — Toulouse, France